

Task 33: Design Circular Queue

Problem Description

Design your implementation of the circular queue. The circular queue is a linear data structure in which the operations are performed based on FIFO (First In First Out) principle and the last position is connected back to the first position to make a circle.

Java Solution

```
class MyCircularQueue {
    int[] q;
    int head = 0, tail = -1, len = 0;

    public MyCircularQueue(int k) {
        q = new int[k];
    }

    public boolean enqueue(int value) {
        if (!isFull()) {
            tail = (tail + 1) % q.length;
            q[tail] = value;
            len++;
            return true;
        } else return false;
    }

    public boolean dequeue() {
        if (!isEmpty()) {
            head = (head + 1) % q.length;
            len--;
            return true;
        } else return false;
    }

    public int Front() {
        return isEmpty() ? -1 : q[head];
    }

    public int Rear() {
        return isEmpty() ? -1 : q[tail];
    }

    public boolean isEmpty() {
        return len == 0;
    }

    public boolean isFull() {
        return len == q.length;
    }
}
```